

Module 4 — Read 1

Study design, two-group inference, and comparing more than two groups

Module 4, Read 1: Designing and analyzing fair comparisons

Project: [Project 4, Designing Fair Comparisons](#) · Preview: [Read 0](#) · Prerequisite: [Module 3](#) · Next: [Quiz A](#), then the [demo labs](#).

This reading is your self-paced textbook for Module 4. It has three sections. Work one section at a time and use the Check yourself box at the end of each before [Quiz A](#).

Throughout, we use one habit for comparing two groups. When you see a difference between the groups in the data, first ask which of three explanations could produce it: it could be by chance, it could be a real causal effect, or it could be confounding. A randomized controlled trial removes confounding as an explanation (random assignment balances other variables), and a small p-value lets you set aside the by-chance explanation. What is left is the causal explanation.

- Section 1, Study design (including confounders). How groups form, the three explanations for a difference, confounding, DAGs, and what randomization buys you.
- Section 2, Two-group tests and confidence intervals. Pick and run the right test for a two-group comparison, and interpret the p-value and 95 percent confidence interval.
- Section 3, Comparing more than two groups. The same logic extends to three or more groups (ANOVA via `lm`) and to binary outcomes (`glm`). Project 4 stays a two-group comparison, so this section is for awareness and your methods map.

Learning outcomes

After this reading you should be able to:

Section 1: Study design (including confounders)

- Compare randomized controlled trials (RCTs) and observational designs when the goal is a causal claim.
- Explain how units end up in comparison groups in an RCT versus an observational study.
- List the three explanations for an observed difference: by chance, causation, confounding.
- Define a confounder in words and in a DAG, and use subclassification to compare groups within levels of a measured confounder.
- Distinguish random assignment (controls confounding) from random sampling (controls bias when generalizing).

Section 2: Two-group tests and confidence intervals

- Use variable types to pick `t.test()` for two independent numerical groups or `prop.test()` / `chisq.test()` for binary comparisons, and name the conditions that match the [methods map](#).
- Interpret Welch two-sample t-test output (hypotheses, p-value, 95 percent CI) in context.
- Identify the parameter of interest: a difference in means for a numerical response, or a difference in proportions for a binary response.
- Recognize paired or matched data and use the paired t-test (numerical) or McNemar's test (binary).

Section 3: Comparing more than two groups

- Use one-way ANOVA (`lm()` then `anova()`) for a numerical outcome across three or more groups.
- Use logistic regression (`glm(..., family = binomial)` then `anova(fit, test = "Chisq")`) for a binary outcome across groups.

- Explain why Project 4 keeps to two groups and where these tools fit on the **methods map**.
-

Section 1: Study design (including confounders)

Project 4 connection

Project 4 asks you to design a fair comparison between two groups. This section gives you the vocabulary: EV, RV, RCT vs observational, confounding, and what a p-value can and cannot tell you about causation.

Causal questions, explanatory and response variables

A causal question asks about the effect of something. Name two roles:

Role	Other names	Meaning
Response (RV)	outcome, dependent	The effect you want to compare.
Explanatory (EV)	treatment, exposure	The potential cause or group label.

When writing a causal research question, it is best practice to:

1. Define variables precisely. State how each variable is measured and any cutoffs (for example pass/fail, deprived vs unrestricted sleep, income in dollars). Vague labels make design and analysis impossible.
2. Identify the observational units. The unit is the entity that gets a value for each variable, often one person at one time point. Your dataset is usually one row per observational unit.
3. Match the summary to the variable types. If the response is numerical, summarize the two groups with means (or medians) and a Welch two-sample t-test. If the response is binary (or reduced to two categories), compare proportions or rates with `prop.test` / `chisq.test`. You will do this in Section 2.

The three explanations for a difference

Suppose the two groups in your data differ. Before claiming anything, list the three explanations:

1. By chance. Random variation alone can produce a gap, even when the groups are really the same. Even fair coins give different numbers of heads in 100 flips.
2. Causation. The explanatory variable really does change the response.
3. Confounding. A third variable influences both the explanatory and response variables and mixes up their relationship, making a fair comparison hard.

A hypothesis test and its p-value address explanation 1 only. A small p-value lets you set aside the by-chance explanation. It does not, by itself, help you choose between causation and confounding. That is the job of study design.

Two design tools tell the explanations apart:

- A randomized controlled trial uses random assignment, which balances other variables across the groups on average and so removes confounding as an explanation. In an RCT only two explanations remain, by chance and causation, and a small p-value points to causation.
- An observational study leaves all three explanations on the table, so confounding must be reasoned about with DAGs and addressed with subclassification when a confounder is measured.

Students often confuse random assignment with random sampling:

- Random assignment decides which group each unit goes into within the sample. It is used to balance confounders so a comparison between groups is fairer.
- Random sampling decides which individuals are selected from a population. It is used to reduce bias when generalizing study results back to a larger population.

Which university is better? (a confounding story)

Suppose a state has two polytechnic universities and the Chancellor's Office wants to know which one does a better job teaching calculus to incoming freshmen.

This example is a modification of a story in Pearl, Judea, and Dana Mackenzie, The Book of Why: The New Science of Cause and Effect (Basic Books, 2018).

Crude pass rates. Both universities have a calculus class of 200 students:

	Passed calculus
Timber Ridge	150/200 (75%)
Wild Horse	154/200 (77%)

Wild Horse has more students who passed (154 vs 150). But these are observational groups: students chose a university; the Chancellor did not randomly assign them. A higher crude rate does not by itself answer the causal question.

If we know who took AP Calculus in high school:

	AP calculus	No AP calculus	Combined
Timber Ridge	45/50 (90%)	105/150 (70%)	150/200 (75%)
Wild Horse	136/170 (80%)	18/30 (60%)	154/200 (77%)

Each cell shows number who passed / number of students in that cell. Read down each column (holding preparation fixed):

- Among students without AP calculus, Timber Ridge has the higher pass rate (70% vs 60%).
- Among students with AP calculus, Timber Ridge again has the higher pass rate (90% vs 80%).

So within each preparation level, Timber Ridge does better. The crude comparison reversed once we accounted for AP background. This reversal is Simpson's paradox, driven by a confounder.

Confounding variable (definition)

A confounder (for the effect of explanatory variable X on response Y) is a variable that influences both X and Y .

- AP calculus confounds university and passing: it influences both where students enroll and how likely they are to pass.
- Nightlight and myopia: parental myopia or genetics may confound nightlight use and child myopia.

DAGs and subclassification

A causal directed acyclic graph (DAG) is a picture of assumed causal paths. Nodes are variables. A directed edge $A \rightarrow B$ means "A helps cause B." Acyclic means no directed cycles. The defining picture of a confounder is two arrows leaving it: one arrow goes from the confounder to the explanatory group, and a second arrow goes from the confounder to the outcome. A third arrow goes from the explanatory group to the outcome (the effect you want to measure).

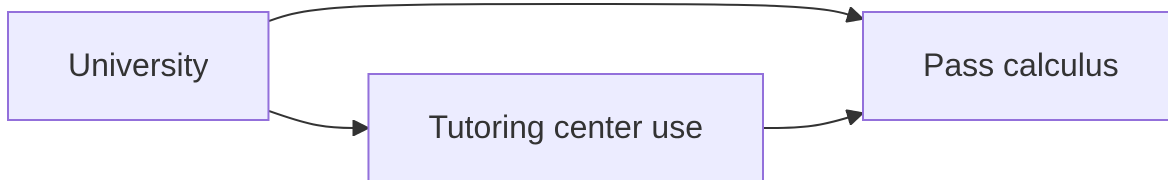
A backdoor path is a non-causal route that leaks association into the crude X to Y link, for example $U \leftarrow A \rightarrow P$ where U is university, P is passing, and A is AP calculus. Conditioning on A (comparing universities within AP and no-AP subtables) blocks that path. This is called subclassification (or stratification).

```
flowchart LR
  A[AP calculus in HS] --> U[University]
  A --> P[Pass calculus]
  U --> P
```



Not everything should be adjusted for. Suppose the campus tutoring center is part of what each university provides. Then tutoring is downstream of university, it lies on the causal path from university to passing, not on a backdoor path. Adjusting for it would remove part of the very effect you want to measure.

```
flowchart LR
  U2[University] --> T[Tutoring center use]
  U2 --> P2[Pass calculus]
  T --> P2
```



So under the AP DAG, the AP vs no-AP subclassification is the appropriate fairer comparison; the tutoring breakdown is not the same kind of adjustment. You can trace backdoor paths by hand or use [DAGitty](#) to suggest adjustment sets under an assumed graph.

Do you believe the diagram? Before adjusting, ask whether you included every common cause linking university, passing, and the other measured variables. Residual confounding from unmeasured variables (motivation, SES) is always possible. This is exactly why real policy questions need richer datasets, and why an RCT, when feasible, is so valuable.

Seeing confounding in data (the three arrows)

A DAG with a confounder has three arrows: confounder to group, confounder to outcome, and group to outcome. Each arrow shows up as an association you can plot and summarize with Module 3 tools. Below we simulate a treatment comparison where age group is a confounder, then look at all three arrows. Older patients are sicker (lower outcome) and, in the observational data, are more likely to receive Treatment B.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}
library(tidyverse)

set.seed(2024)
```

```

N <- 300
# Confounder: age group (younger / older), about half and half
age_group <- sample(c("younger", "older"), N, replace = TRUE, prob = c(0.5, 0.5))

# Arrow 1 (confounder -> group): older patients are more likely to get Treatment B
p_B <- ifelse(age_group == "older", 0.75, 0.25)
treatment <- ifelse(rbinom(N, 1, p_B) == 1, "B", "A")

# Arrow 2 (confounder -> outcome): older patients have lower outcomes.
# Arrow 3 (group -> outcome): Treatment B has a small TRUE benefit of +3.
outcome <- 70 - 12 * (age_group == "older") + 3 * (treatment == "B") + rnorm(N, 0, 6)

conf <- tibble(
  age_group = factor(age_group, levels = c("younger", "older")),
  treatment = factor(treatment, levels = c("A", "B")),
  outcome = outcome
)
glimpse(conf)

```

Rows: 300

Columns: 3

```

$ age_group <fct> younger, older, younger, younger, older, younger, older, old~
$ treatment <fct> A, B, A, A, A, A, A, B, A, A, A, A, A, A, B, A, B, B, B, B, ~
$ outcome <dbl> 74.26722, 57.20574, 68.09254, 76.40501, 67.55353, 72.69615, ~

```

Arrow 1, confounder to group. The age mix is different in the two treatment groups, so the groups are not comparable at baseline.

```
conf |> count(treatment, age_group)
```

A tibble: 4 x 3

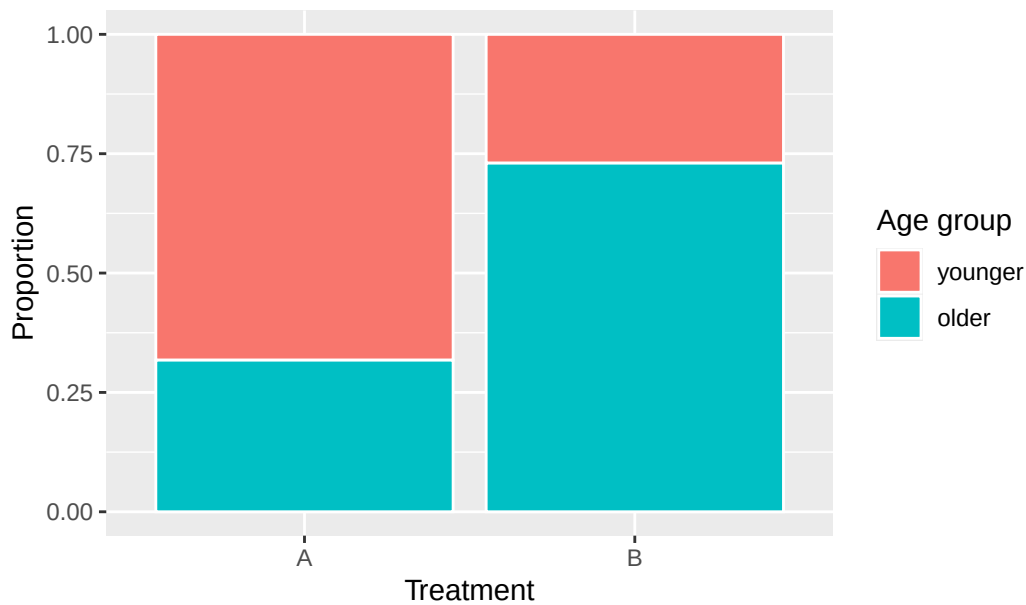
	treatment	age_group	n
	<fct>	<fct>	<int>
1	A	younger	101
2	A	older	47
3	B	younger	41
4	B	older	111

```

conf |>
  count(treatment, age_group) |>
  ggplot(aes(x = treatment, y = n, fill = age_group)) +
  geom_col(position = "fill", color = "white") +
  labs(title = "Arrow 1: age mix differs by treatment group (confounder to group)",
       x = "Treatment", y = "Proportion", fill = "Age group")

```

Arrow 1: age mix differs by treatment group (confounder to grc)



Arrow 2, confounder to outcome. Ignoring treatment, older patients have lower outcomes.

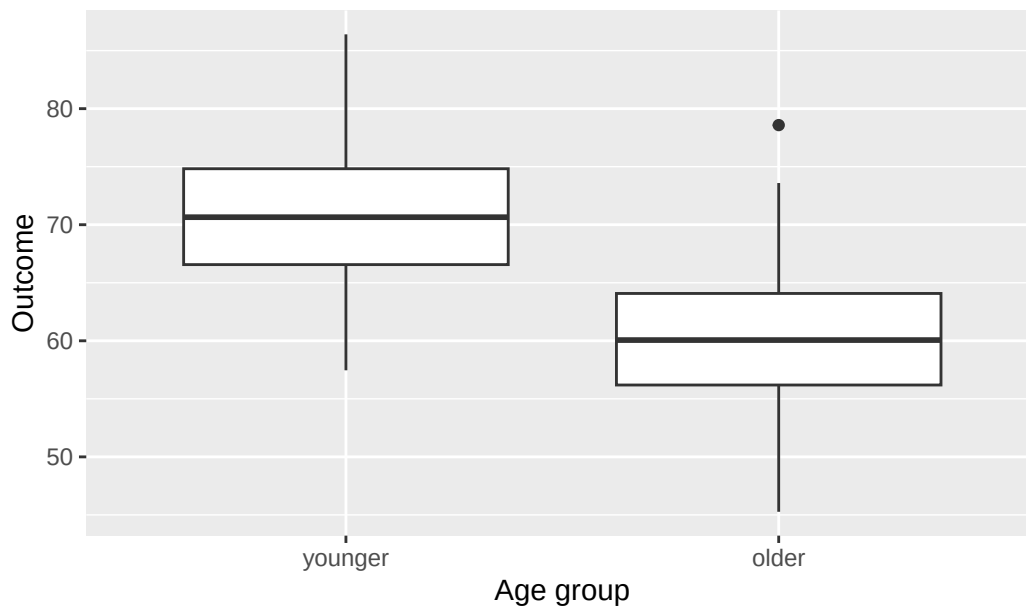
```
conf |>
  group_by(age_group) |>
  summarize(n = n(), mean_outcome = mean(outcome), sd_outcome = sd(outcome))
```

A tibble: 2 x 4

age_group	n	mean_outcome	sd_outcome
<fct>	<int>	<dbl>	<dbl>
1 younger	142	70.7	6.15
2 older	158	60.4	5.88

```
ggplot(conf, aes(x = age_group, y = outcome)) +
  geom_boxplot() +
  labs(title = "Arrow 2: outcome depends on age group (confounder to outcome)",
       x = "Age group", y = "Outcome")
```


Arrow 2: outcome depends on age group (confounder to outcome)



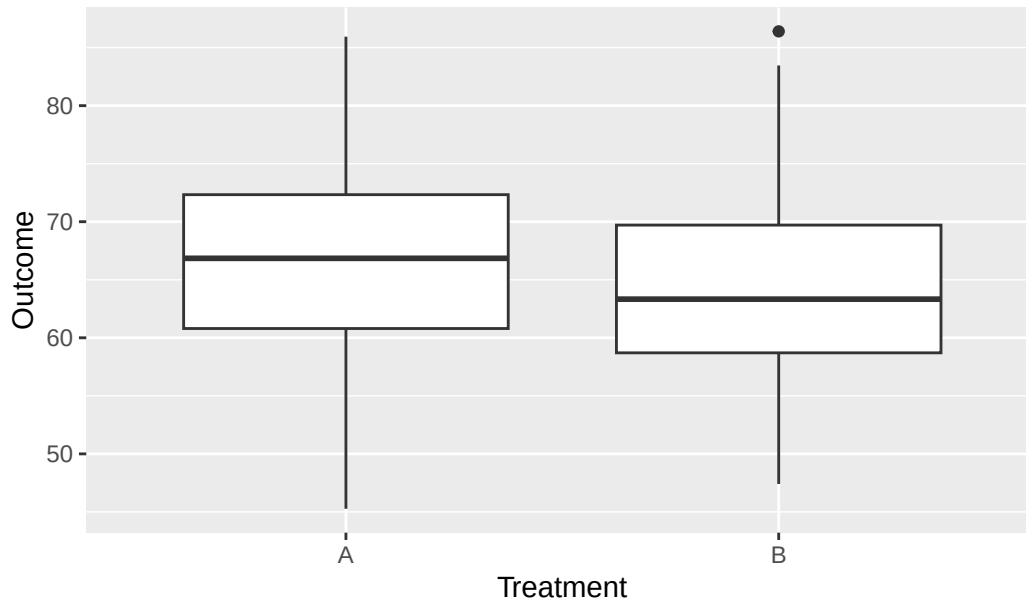
Arrow 3, the crude group-to-outcome comparison. Because older (sicker) patients were more likely to get B, the crude comparison is biased against B, even though B truly helps a little.

```
conf |>
  group_by(treatment) |>
  summarize(n = n(), mean_outcome = mean(outcome), sd_outcome = sd(outcome))
```

```
# A tibble: 2 x 4
  treatment      n mean_outcome sd_outcome
  <fct>      <int>      <dbl>    <dbl>
1 A          148        66.3      8.25
2 B          152        64.3      7.48
```

```
ggplot(conf, aes(x = treatment, y = outcome)) +
  geom_boxplot() +
  labs(title = "Arrow 3: crude outcome by treatment (confounded comparison)",
       x = "Treatment", y = "Outcome")
```

Arrow 3: crude outcome by treatment (confounded comparison)



```
t.test(outcome ~ treatment, data = conf, var.equal = FALSE)
```

Welch Two Sample t-test

data: outcome by treatment

t = 2.2036, df = 293.41, p-value = 0.02833

alternative hypothesis: true difference in means between group A and group B is not equal to 0

95 percent confidence interval:

0.2143178 3.7964894

sample estimates:

mean in group A mean in group B

66.27307

64.26767

The crude p-value can be small while pointing the wrong way: B looks worse than A because of the age imbalance, not because B is harmful. The test removed the by-chance explanation, but confounding remains.

Remove one arrow: randomize the treatment

Now keep the same people and the same age-to-outcome arrow, but assign treatment at random. Random assignment erases arrow 1 (the confounder no longer drives group membership), so age is balanced across groups and the crude comparison becomes fair.

```
set.seed(2025)
treatment_rct <- sample(c("A", "B"), N, replace = TRUE) # random assignment, ignores age
outcome_rct <- 70 - 12 * (age_group == "older") + 3 * (treatment_rct == "B") + rnorm(N, 0, 6)

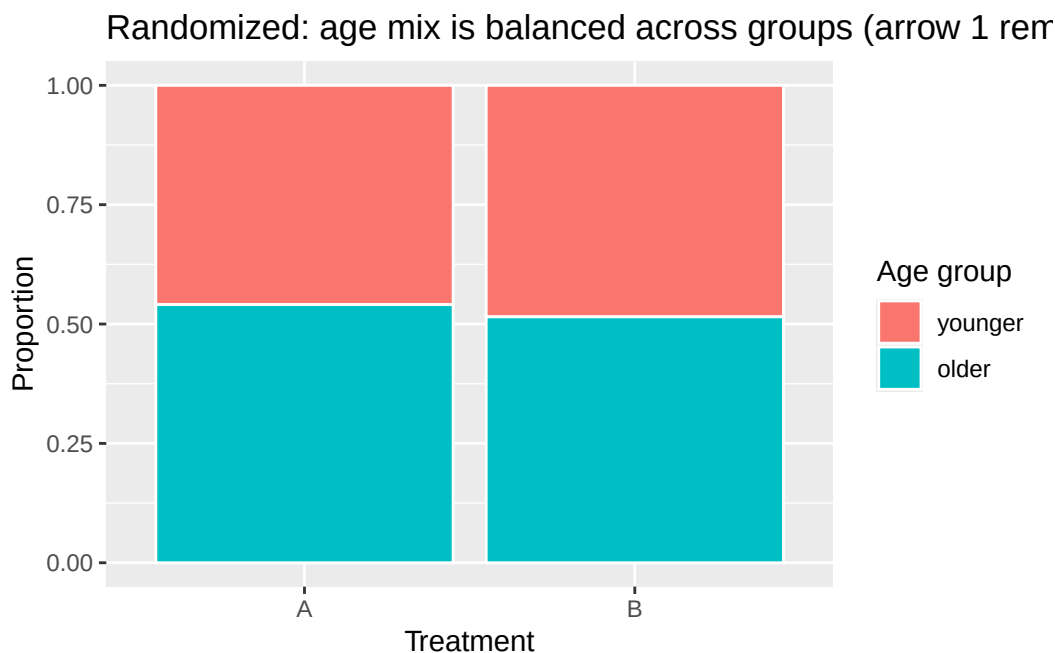
rct <- tibble(
  age_group = factor(age_group, levels = c("younger", "older")),
  treatment = factor(treatment_rct, levels = c("A", "B")),
  outcome = outcome_rct
)

# Arrow 1 is now gone: age mix is about equal in both groups
```

```
rct |> count(treatment, age_group)
```

```
# A tibble: 4 x 3
  treatment age_group     n
  <fct>      <fct>   <int>
1 A        younger    62
2 A        older     73
3 B        younger    80
4 B        older     85
```

```
rct |>
  count(treatment, age_group) |>
  ggplot(aes(x = treatment, y = n, fill = age_group)) +
  geom_col(position = "fill", color = "white") +
  labs(title = "Randomized: age mix is balanced across groups (arrow 1 removed)",
       x = "Treatment", y = "Proportion", fill = "Age group")
```



```
# Crude comparison now recovers the true +3 benefit of B
rct |>
  group_by(treatment) |>
  summarize(n = n(), mean_outcome = mean(outcome), sd_outcome = sd(outcome))
```

```
# A tibble: 2 x 4
  treatment     n mean_outcome sd_outcome
  <fct>      <int>      <dbl>    <dbl>
1 A         135      63.8      7.89
2 B         165      67.1      8.37
```

```
t.test(outcome ~ treatment, data = rct, var.equal = FALSE)
```

Welch Two Sample t-test

data: outcome by treatment

```

t = -3.5367, df = 292.03, p-value = 0.0004711
alternative hypothesis: true difference in means between group A and group B is not equal to 0
95 percent confidence interval:
 -5.181136 -1.476314
sample estimates:
mean in group A mean in group B
    63.75346      67.08218

```

With arrow 1 removed, arrow 2 (age to outcome) still exists, but age is spread evenly across the groups, so it no longer biases the comparison. The crude difference in means now estimates the real effect of treatment, and a small p-value supports a causal conclusion.

The randomized controlled trial (RCT)

A Randomized Controlled Trial (RCT) randomly assigns units to two or more groups. The assigned group becomes the explanatory variable, and a response is measured for each unit.

Internal validity is how well a study rules out alternative explanations for a difference between its groups. Random assignment balances both measured and unmeasured confounders on average, so it removes confounding as a competing explanation. That leaves only causation and by chance, and Section 2's tests handle the by-chance explanation.

When an RCT is impossible or unethical (you cannot randomly assign people to smoke for 20 years), you are back to an observational design and must reason about confounding with DAGs and subclassification.

Check yourself, Section 1 (one per key term, plus scenarios)

Key terms

1. Explanatory variable (EV): define it.
2. Response variable (RV): define it.
3. Observational study: define it.
4. Randomized controlled trial (RCT): define it and give its key feature.
5. Random assignment: what does it accomplish?
6. Random sampling: what does it accomplish, and how is it different from random assignment?
7. By chance: what does this explanation mean for a difference between two groups?
8. Causation: what would have to be true for this explanation to be the right one?
9. Confounding: define it as an explanation for a group difference.
10. Confounder: define it and name the two arrows that define it in a DAG.
11. DAG: what do nodes and directed edges represent?
12. Backdoor path: what is it, and how do you block it?
13. Subclassification (stratification): what is it, and what is its limitation?
14. Internal validity: define it.
15. Simpson's paradox: describe what happened in the university example.

Scenarios

16. A study compares smokers and non-smokers on lung cancer rates. RCT or observational? Why?
17. In the three-arrows simulation, why did the crude treatment comparison point the wrong way before randomization?
18. After randomizing the treatment, age still affects the outcome. Why does it no longer bias the comparison?

Answers, Section 1

1. The variable thought to cause or predict the response (treatment, exposure, grouping factor).
 2. The variable we measure to see whether it changes with the EV.
 3. Researchers record data as-is; group membership is self-selected.
 4. An experiment with random assignment of units to groups.
 5. Balances measured and unmeasured confounders across groups on average.
 6. Random sampling selects individuals from a population (reduces generalization bias); random assignment splits a sample into groups (controls confounding).
 7. Random variation alone could produce the gap even if groups are equal.
 8. The EV truly changes the RV, with chance and confounding ruled out.
 9. A third variable produces or distorts the difference.
 10. A common cause of EV and RV; arrows go from the confounder to the group and to the outcome.
 11. Nodes are variables; an edge from A to B means A helps cause B.
 12. A non-causal EV-to-RV path through a common cause; block it by subclassifying on the confounder.
 13. Compare within confounder levels; only handles measured confounders.
 14. How well a study rules out alternative explanations within its groups.
 15. Crude rates favored Wild Horse, but Timber Ridge was higher within each AP level.
 16. Observational; smoking cannot be randomly assigned.
 17. Older patients were more likely to get B, so the crude difference was confounded by age.
 18. Randomization balances age across groups, so it no longer biases the comparison.
-

Section 2: Two-group tests and confidence intervals

Design (Section 1) decides whether a difference can be called causal. This section handles the by-chance explanation. A hypothesis test asks: if the two groups were really the same, how surprising is a difference this large? The p-value answers that. A 95 percent confidence interval estimates how big the difference is, with uncertainty.

The first decision is the parameter of interest, and it follows from the response variable type:

- Numerical response (a measured number per unit): the parameter is a difference in means, $\mu_1 - \mu_2$. Summarize with group means and standard deviations; compare with a Welch two-sample t-test.
- Binary response (each unit is one of two categories): the parameter is a difference in proportions, $p_1 - p_2$. Summarize with group proportions (rates); compare with `prop.test()` or, equivalently, a chi-square test of independence.

Methods table for two groups

Response variable	Plot (Module 3)	Summaries	Test	CI	Conditions (match the methods map)
Numerical	Boxplot per group	Group n , mean, SD	<code>t.test(y ~ group, var.equal = FALSE)</code>	two-sample t-interval from same call	Two samples or random assignment; independent observations; each group large or roughly normal/symmetric
Binary	100% stacked bar	Group counts and proportions	<code>prop.test(tab)</code> or <code>chisq.test(tab)</code>	two-proportion CI from <code>prop.test</code>	Independent observations; all expected counts at least 5 (for the z/chi-square approximation)

For paired or matched data, use the paired versions instead (covered below).

Numerical response: Welch two-sample t-test

Scenario. In a sleep study, 30 volunteers are randomly assigned to a deprived group (4 hours) or a rested group (8 hours). The next morning each completes a reaction-time task (lower is better, in milliseconds). Because assignment was random, this is an RCT, so confounding is removed by design and we only need to address the by-chance explanation.

Parameter of interest: the difference in mean reaction time, $\mu_{\text{deprived}} - \mu_{\text{rested}}$.

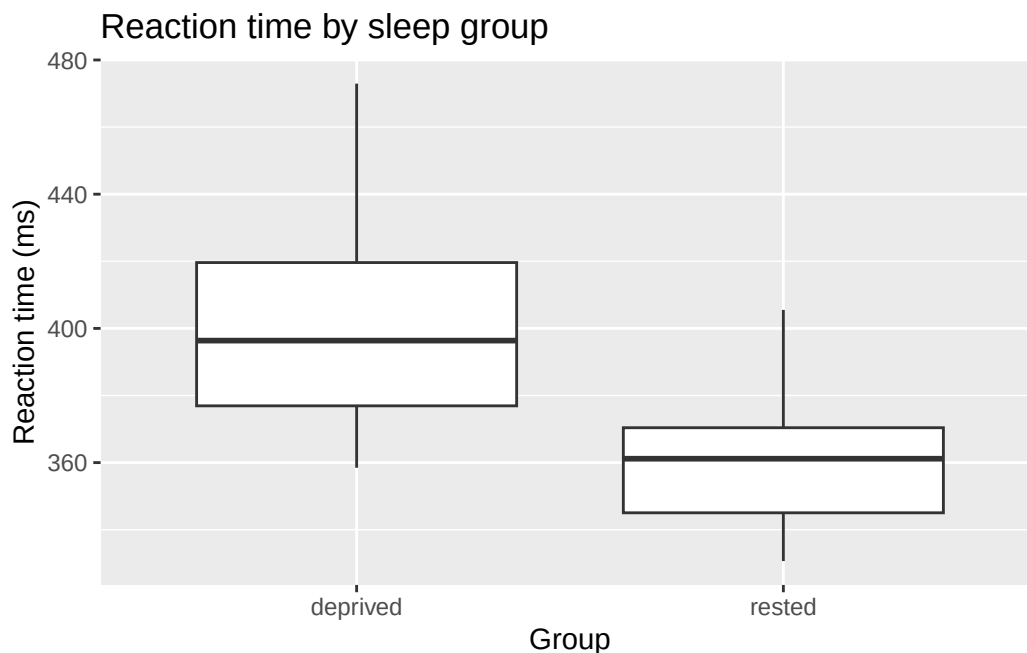
```
library(tidyverse)
set.seed(11)
sleep_rt <- tibble(
  group = rep(c("deprived", "rested"), each = 15),
  rt = c(rnorm(15, mean = 420, sd = 40),
        rnorm(15, mean = 370, sd = 40))
)

# Module 3 summaries: n, mean, SD per group
```

```
sleep_rt |>
  group_by(group) |>
  summarize(n = n(), mean_rt = mean(rt), sd_rt = sd(rt))
```

```
# A tibble: 2 x 4
  group      n mean_rt sd_rt
  <chr>  <int>   <dbl> <dbl>
1 deprived    15    403.   36.6
2 rested     15    361.   19.3
```

```
# Module 3 plot: boxplot per group
ggplot(sleep_rt, aes(x = group, y = rt)) +
  geom_boxplot() +
  labs(title = "Reaction time by sleep group", x = "Group", y = "Reaction time (ms)")
```



```
t.test(rt ~ group, data = sleep_rt, var.equal = FALSE)
```

Welch Two Sample t-test

```
data:  rt by group
t = 3.9333, df = 21.223, p-value = 0.0007495
alternative hypothesis: true difference in means between group deprived and group rested is not equal to 0
95 percent confidence interval:
 19.82771 64.25533
sample estimates:
mean in group deprived    mean in group rested
      402.8768             360.8352
```

How to read the output:

- Hypotheses: $H_0 : \mu_{\text{deprived}} - \mu_{\text{rested}} = 0$ versus $H_a : \mu_{\text{deprived}} - \mu_{\text{rested}} \neq 0$.
- p-value: the probability of a difference at least this large if the two means were equal. A small p-value (say below 0.05) lets you set aside the by-chance explanation.

- 95 percent CI: a plausible range for $\mu_{\text{deprived}} - \mu_{\text{rested}}$. If the whole interval is above 0, deprived reaction times are slower. If the interval includes 0, the data are consistent with no difference.
- Because this is an RCT, a small p-value supports a causal reading: deprivation increased reaction time.

Conditions (from the methods map): independent observations and random assignment hold by design; with 15 per group and roughly symmetric data, the t-approximation is reasonable. Always check the boxplot for strong skew or extreme outliers first.

Binary response: two proportions

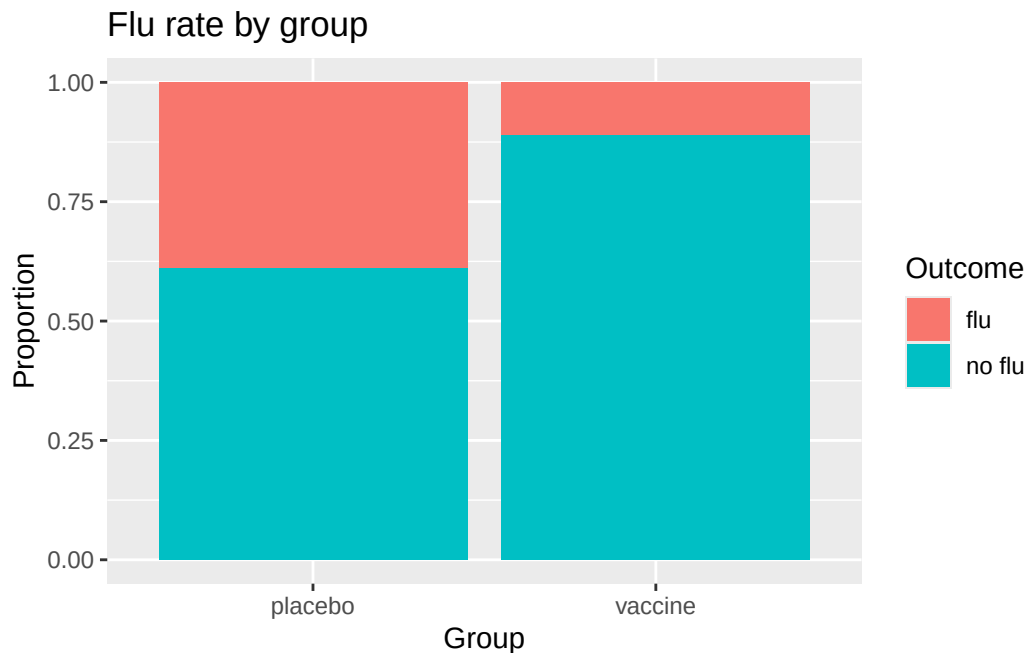
Scenario. A vaccine trial randomly assigns 200 volunteers to vaccine or placebo and records whether each person caught the flu that season (yes/no). The response is binary, so the parameter of interest is the difference in infection proportions, $p_{\text{vaccine}} - p_{\text{placebo}}$.

```
library(tidyverse)
set.seed(7)
vax <- tibble(
  group = rep(c("vaccine", "placebo"), each = 100),
  flu = c(rbinom(100, 1, 0.10), rbinom(100, 1, 0.30))
) |>
  mutate(flu = factor(ifelse(flu == 1, "flu", "no flu"), levels = c("flu", "no flu")))

# Module 3 summaries: counts and infection rate per group
vax |>
  group_by(group) |>
  summarize(n = n(), flu_count = sum(flu == "flu"), flu_rate = mean(flu == "flu"))
```

```
# A tibble: 2 x 4
  group      n flu_count flu_rate
  <chr> <int>   <int>   <dbl>
1 placebo  100     39     0.39
2 vaccine  100     11     0.11
```

```
# Module 3 plot: 100% stacked bar
ggplot(vax, aes(x = group, fill = flu)) +
  geom_bar(position = "fill") +
  labs(title = "Flu rate by group", x = "Group", y = "Proportion", fill = "Outcome")
```

```
tab <- table(vax$group, vax$flu)
tab
```

```
      flu no flu
placebo  39   61
vaccine  11   89
```

```
prop.test(tab)      # difference in proportions, with 95% CI
```

2-sample test for equality of proportions with continuity correction

```
data:  tab
X-squared = 19.44, df = 1, p-value = 1.038e-05
alternative hypothesis: two.sided
95 percent confidence interval:
 0.1564235 0.4035765
sample estimates:
prop 1 prop 2
 0.39   0.11
```

```
chisq.test(tab)      # equivalent test of independence for a 2x2 table
```

Pearson's Chi-squared test with Yates' continuity correction

```
data:  tab
X-squared = 19.44, df = 1, p-value = 1.038e-05
```

`prop.test()` reports the two group proportions, a p-value, and a 95 percent CI for $p_1 - p_2$. `chisq.test()` gives the same conclusion as a test of independence; use `chisq.test(tab)$expected` to confirm the condition that all expected counts are at least 5. For a 2x2 table the two tests agree.

i Inference reference box: two categorical variables

For a 2x2 comparison of a binary outcome between two groups, `prop.test(tab)` and `chisq.test(tab)` test the same null (no association between group and outcome) and reach the same conclusion. Report `prop.test()` when you also want the CI for $p_1 - p_2$. Condition: independent observations and all expected counts at least 5.

Paired or matched comparisons

Sometimes the two measurements are linked: the same unit measured twice (before/after), or units matched into pairs. Then the observations are not independent across groups, and the independent-samples tests above do not apply. Use the paired versions, which analyze the within-pair differences.

Numerical outcome, paired t-test. Example: blood pressure for 12 patients, before and after a program. Test the mean of the before-minus-after differences.

```
library(tidyverse)
set.seed(3)
bp <- tibble(
  patient = 1:12,
  before = rnorm(12, 150, 10),
  after = rnorm(12, 142, 10)
)

bp |> summarize(mean_before = mean(before), mean_after = mean(after),
  mean_change = mean(before - after))
```

```
# A tibble: 1 x 3
  mean_before mean_after mean_change
    <dbl>         <dbl>         <dbl>
1    148.         139.           9.37
```

```
t.test(bp$before, bp$after, paired = TRUE)
```

Paired t-test

```
data: bp$before and bp$after
t = 3.298, df = 11, p-value = 0.007103
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 3.115704 15.617766
sample estimates:
mean difference
 9.366735
```

Binary outcome, McNemar's test. Example: 80 people rated a policy support yes/no before and after a debate. The 2x2 table cross-classifies each person's before and after answers; McNemar's test uses only the people who switched.

```
support <- matrix(c(20, 25, 10, 25),
  nrow = 2, byrow = TRUE,
  dimnames = list(before = c("yes", "no"),
    after = c("yes", "no")))
support
```

after

```
before yes no
  yes  20 25
  no   10 25
```

```
mcnemar.test(support)
```

McNemar's Chi-squared test with continuity correction

```
data: support
```

```
McNemar's chi-squared = 5.6, df = 1, p-value = 0.01796
```

Independence vs paired is a design question, not a data question. Ask: was each value in group 1 produced by the same unit (or a matched partner) as a specific value in group 2? If yes, the data are paired and you use the paired test; if the two groups are separate, unrelated units, use the independent-samples test.

Ruling out chance without a formula (optional)

You can also address the by-chance explanation by simulation: shuffle the group labels many times, recompute the difference, and see how often a difference as large as the observed one appears by chance alone. The free [Rossman-Chance applets](#) let you do this for both means and proportions, and the demo labs show the same idea in R with `set.seed()` and resampling.

Check yourself, Section 2 (key terms, scenarios, paired-or-not)

Key terms

1. By-chance explanation: what does a hypothesis test do about it?
2. p-value: state its meaning in one sentence.
3. Confidence interval (95 percent): what does it estimate, and what does it mean if it contains 0?
4. Parameter of interest: how do you decide between a difference in means and a difference in proportions?
5. Welch two-sample t-test: when do you use it, and what are its conditions?
6. Two-proportion test (`prop.test`): when do you use it, and what are its conditions?
7. Chi-square test of independence: what null does it test, and what is the expected-count condition?
8. Paired t-test: when do you use it instead of the two-sample t-test?
9. McNemar's test: when do you use it?

Choose the test/CI and state its conditions (use the methods map)

10. A clinic randomly assigns 60 patients to drug or placebo and records cholesterol (mg/dL) after 8 weeks. Identify the design, the parameter of interest, the test, and its conditions.
11. A survey records whether each of 300 adults exercises (yes/no) and whether they own a dog (yes/no), and asks if dog owners exercise more. Identify the design, parameter, test, and conditions.
12. Researchers measure each of 25 runners' resting heart rate before and after a 10-week program. Identify the design, parameter, test, and conditions.

Paired or not?

13. Group 1 is 40 men and group 2 is 40 different women; compare mean income. Paired or independent?
14. Each of 30 students takes a pre-test and a post-test; compare mean scores. Paired or independent?
15. 50 patients are each measured on the left and right arm; compare mean blood pressure. Paired or independent?

Answers, Section 2

1. It measures how surprising the difference is under the null and lets you set aside chance when small.
 2. Probability of a difference at least this large if the null (no difference) is true.
 3. A plausible range for the true difference; containing 0 means consistent with no difference.
 4. Numerical response gives a difference in means; binary response gives a difference in proportions.
 5. Two independent numerical groups; conditions: independence/random assignment, each group large or roughly symmetric.
 6. Two independent groups, binary outcome; conditions: independence, expected counts at least 5.
 7. Null: group and outcome independent; condition: independence and expected counts at least 5.
 8. When measurements are linked (before/after or matched pairs).
 9. Binary outcome measured twice on the same units (or matched pairs).
 10. RCT; difference in means; Welch t-test + t-interval; independence/random assignment, each group large or roughly symmetric.
 11. Observational; difference in proportions; prop.test/chisq.test; independence, expected counts at least 5; association not causation.
 12. Paired; mean difference; paired t-test + t-interval; differences roughly normal or large n.
 13. Independent. 14. Paired. 15. Paired.
-

Section 3: Comparing more than two groups

i Scope for Project 4

Project 4 stays a single two-group comparison (one binary explanatory contrast). This section is for awareness and your methods map, so you recognize the right tool when there are three or more groups.

The three-explanations logic does not change with more groups. Design still decides whether a difference can be causal, and a test still handles the by-chance explanation. Only the test changes, because comparing three or more group means at once with many t-tests inflates the chance of a false positive.

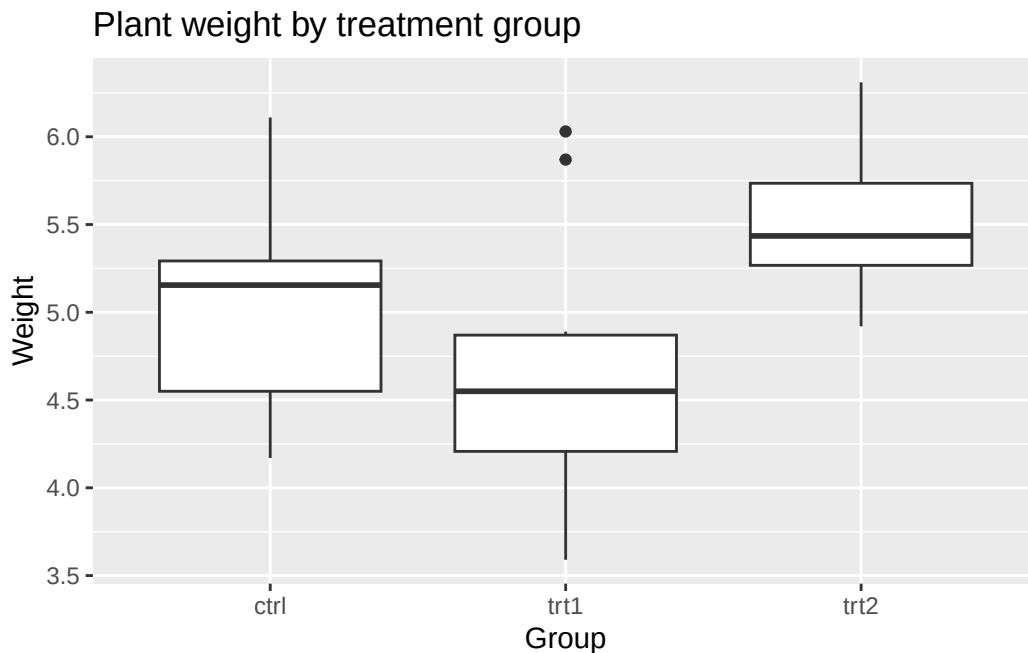
Numerical outcome, three or more groups: one-way ANOVA

One-way ANOVA tests whether all group means are equal: $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$. Fit a linear model with `lm()` and read the overall test with `anova()`.

```
library(tidyverse)
PlantGrowth |>
  group_by(group) |>
  summarize(n = n(), mean_weight = mean(weight), sd_weight = sd(weight))
```

```
# A tibble: 3 x 4
  group      n mean_weight sd_weight
<fct> <int>      <dbl>      <dbl>
1 ctrl      10      5.03      0.583
2 trt1      10      4.66      0.794
3 trt2      10      5.53      0.443
```

```
ggplot(PlantGrowth, aes(x = group, y = weight)) +
  geom_boxplot() +
  labs(title = "Plant weight by treatment group", x = "Group", y = "Weight")
```



```
fit <- lm(weight ~ group, data = PlantGrowth)
anova(fit)
```

Analysis of Variance Table

Response: weight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	3.7663	1.8832	4.8461	0.01591 *
Residuals	27	10.4921	0.3886		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The `anova()` p-value tests whether any group mean differs. Conditions (from the methods map): independent observations, roughly normal within groups, and similar spread across groups. A small p-value says at least one group differs; follow-up comparisons say which.

Binary outcome across groups: logistic regression

When the response variable is binary and you compare across groups, fit a logistic regression with `glm(..., family = binomial)` and test the group effect with `anova(fit, test = "Chisq")`.

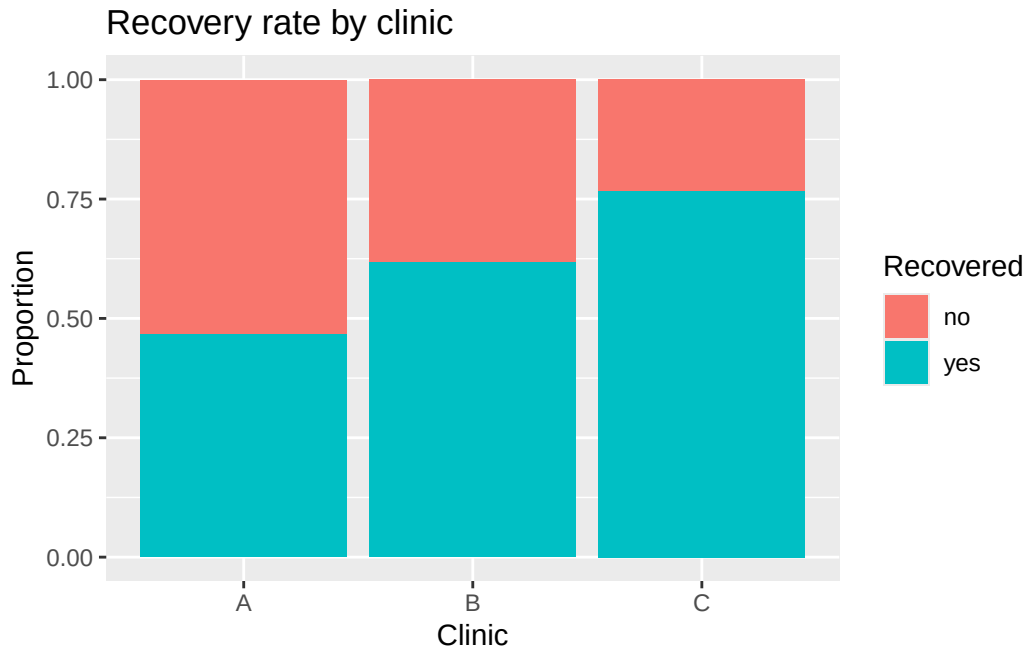
```
set.seed(5)
clinic <- tibble(
  clinic = factor(rep(c("A", "B", "C"), each = 60)),
  recovered = c(rbinom(60, 1, 0.5), rbinom(60, 1, 0.65), rbinom(60, 1, 0.8))
)

clinic |>
  group_by(clinic) |>
  summarize(n = n(), recovered_rate = mean(recovered))
```

```
# A tibble: 3 x 3
  clinic      n recovered_rate
```

	<fct>	<int>	<dbl>
1	A	60	0.467
2	B	60	0.617
3	C	60	0.767

```
ggplot(clinic, aes(x = clinic, fill = factor(recovered, labels = c("no", "yes")))) +
  geom_bar(position = "fill") +
  labs(title = "Recovery rate by clinic", x = "Clinic", y = "Proportion", fill = "Recovered")
```



```
fit_glm <- glm(recovered ~ clinic, data = clinic, family = binomial)
anova(fit_glm, test = "Chisq")
```

Analysis of Deviance Table

Model: binomial, link: logit

Response: recovered

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			179	239.64	
clinic	2	11.658	177	227.98	0.002941 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Conditions (from the methods map): a binary response, independent observations, and an appropriate model form. The `anova(fit, test = "Chisq")` p-value tests whether recovery probability differs across clinics.

Check yourself, Section 3

1. One-way ANOVA: what null does it test, and what R code produces the overall p-value?
2. Why not just run many two-sample t-tests across several groups?
3. State the ANOVA conditions from the methods map.
4. Logistic regression: when do you use `glm(..., family = binomial)` to compare groups, and how do you get the group p-value?
5. A study compares mean recovery time across four hospitals (numerical outcome). Which tool, and what plot would you show first?

Answers, Section 3

1. All group means equal; `lm(y ~ group)` then `anova(fit)`.
 2. Many pairwise tests inflate the false-positive rate; ANOVA is one overall test.
 3. Independent observations, roughly normal within groups, similar spread.
 4. Binary outcome across groups; group p-value from `anova(fit, test = "Chisq")`.
 5. One-way ANOVA; boxplot per hospital first.
-

Where to go next

- Take [Quiz A](#) to check definitions, study-design identification, and method choice.
- Work the [demo labs](#): Demo 1 simulates data by study design and compares groups with Module 3 plots and summaries; Demo 2 runs the tests and confidence intervals from Section 2.
- Review the [methods map](#) so your test choice and stated conditions always match the course reference.
- Start [Project 4](#), keeping to a single two-group comparison.